CASTING PRODUCTS WITH 3D PRINTED MOLDS

Researching the best process for casting silicon molds with a 3D printer, and compiling our findings in an easy-to-use web guide

COMBINING THE VERSATILITY OF 3D PRINTING AND PROPERTIES OF CAST MATERIALS

3D printing allows for incredibly rapid and cost-effective production of objects, but there are significant limitations to the materials that 3D printers can utilize

Casting materials can have much more varied and often valuable material properties (e.g. high flexibility, chemical resistance)

With an optimized process of directly printing a mold to be used for casting, we can get all the benefits from a cast material with the convenience and versatility of 3D printing.

There are a number of potential applications to the combination of these processes – from medial implants to mechanical parts and linkages.

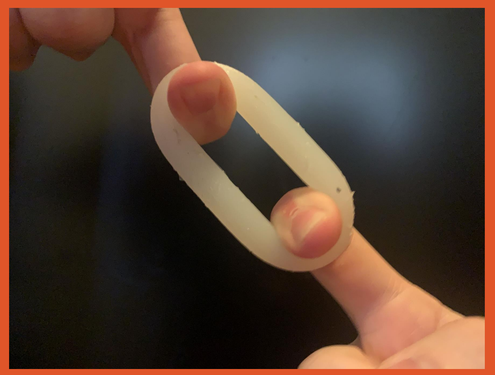


Figure 3. Cast parts can have far greater flexibility than 3D prints

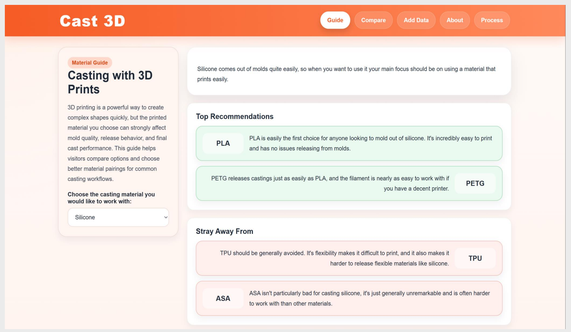


Figure 1. Web guide home page

THE WEB GUIDE

The web guide is designed to be as simple and user friendly as possible, with necessary information available to the user in as little as two clicks

Inspired by the website thistothat.com, our site asks the user what they want to work with, and they are immediately presented with a list of options relevant to them.

The site expands upon this layout by also allowing comparisons of individual tests, and for users to upload test results of their own.

All of the data that was collected by our team is supported by a guide to replicate our results.

THE TESTING PROCESS

- We tested a mix of common and more specialized 3D printing materials to see how they would interact with our set of casting materials
- We selected a small sample of casting materials based on their cured material hardness, as one of the key considerations with casting is releasing casts – which is most affected by material hardness
- We found that the cure rate was a significant factor in casting molds – faster cure rate means more heat, and too much heat will cause a model to fuse to its mold.

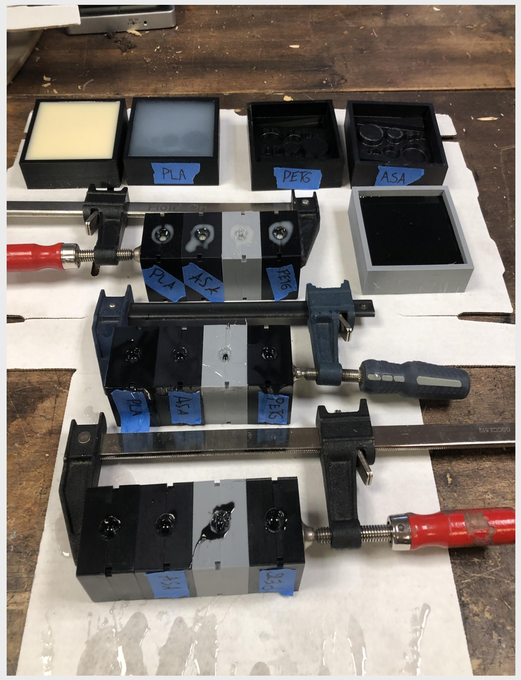


Figure 2. Molds after pouring casting materials

OUR TEAM

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OUR TOOLS

3D printed parts for our project were produced through the OSU 3D printing club. Materials were provided by our project partner, Dean Akin.

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